Industrial Conveyor Automation using PLC with Timer and Safety Interlocks

🏭 PLC-Based Conveyor Motor Control System with Auto, Manual, Timer & Safety Interlocks

---

📌 1. Introduction

In modern industrial automation, conveyor systems are widely used for material handling and production processes. Efficient operation and safety are critical to ensure continuous production and to prevent accidents.

Traditional manual control systems are limited in flexibility and safety. To overcome these challenges, a Programmable Logic Controller (PLC)-based control system is implemented.

This project focuses on designing a conveyor motor control system using GX Works3 from Mitsubishi Electric, integrating manual and automatic modes, timer-based control, and safety interlocks.

🎯 2. Objective

- To design a PLC-based conveyor motor control system
- To implement both manual and automatic modes of operation
- To incorporate timer-based delay in automatic mode
- To ensure safety using emergency stop and overload protection
- To provide visual indication of system status

⚙️ 3. System Description

The system controls a conveyor motor using a PLC. It operates under two modes:

🟢 Manual Mode

In manual mode, the operator directly controls the motor using push buttons. The motor starts immediately when the START button is pressed and stops when the STOP button is pressed.

🔵 Automatic Mode

In automatic mode, a sensor detects the presence of an object on the conveyor. Once detected, a timer delay is initiated. After the delay, the motor starts automatically.

🔴 Safety System

Safety is a critical part of the system. Emergency stop and overload protection are implemented. If any fault occurs, the motor stops immediately regardless of the operating mode.

🔌 4. I/O Table

Device	Address	Description
Start Button	X0	Starts the motor
Stop Button	X1	Stops the motor
Emergency Stop	X2	Emergency shutdown
Overload	X3	Motor protection
Mode Selector	X4	Manual / Auto selection
Sensor	X5	Object detection
Motor	Y0	Conveyor motor
Green Lamp	Y1	Motor running indication
Red Lamp	Y2	Fault indication
Yellow Lamp	Y3	Timer running indication

🧠 5. Control Logic

🔷 Mode Selection

The system selects operation mode based on the selector switch:

- X4 = OFF → Manual Mode
- X4 = ON → Automatic Mode

🔷 Manual Mode Logic

In manual mode:

- The motor starts when the START button is pressed
- A latch mechanism keeps the motor running
- The motor stops when the STOP button is pressed
- Safety conditions must be satisfied for operation

🔷 Automatic Mode Logic

In automatic mode:

- Sensor detects the object
- Timer starts counting
- After timer completion, the motor runs automatically
- Motor stops when the object is no longer detected

🔷 Safety Logic

Safety conditions override all operations:

- Emergency stop (X2) immediately stops the motor
- Overload (X3) protects the motor from damage

⏱️ 6. Timer Explanation

A timer (T0) is used in automatic mode to introduce a delay between object detection and motor operation.

- Timer Type: ON-delay timer
- Preset Value: 3 seconds
- Time Base: 0.1 seconds

This delay ensures controlled movement and proper spacing of products on the conveyor.

💡 7. Indicator System

The system includes indicator lamps for visual feedback:

- 🟢 Green Lamp (Y1): ON when the motor is running
- 🔴 Red Lamp (Y2): ON during fault conditions (Emergency or Overload)
- 🟡 Yellow Lamp (Y3): ON during timer operation (waiting period)

These indicators help the operator understand system status easily.

🧪 8. Testing

The system was tested using the simulation environment of GX Works3.

Test Conditions:

- Manual start and stop operation
- Automatic operation with sensor input
- Timer delay verification
- Emergency stop functionality
- Overload protection
- Indicator lamp operation

All test cases were performed and verified successfully.

📊 9. Results

- The conveyor motor operated correctly in both manual and automatic modes
- Timer delay functioned accurately
- Safety conditions stopped the system instantly
- Indicator lamps displayed correct system status

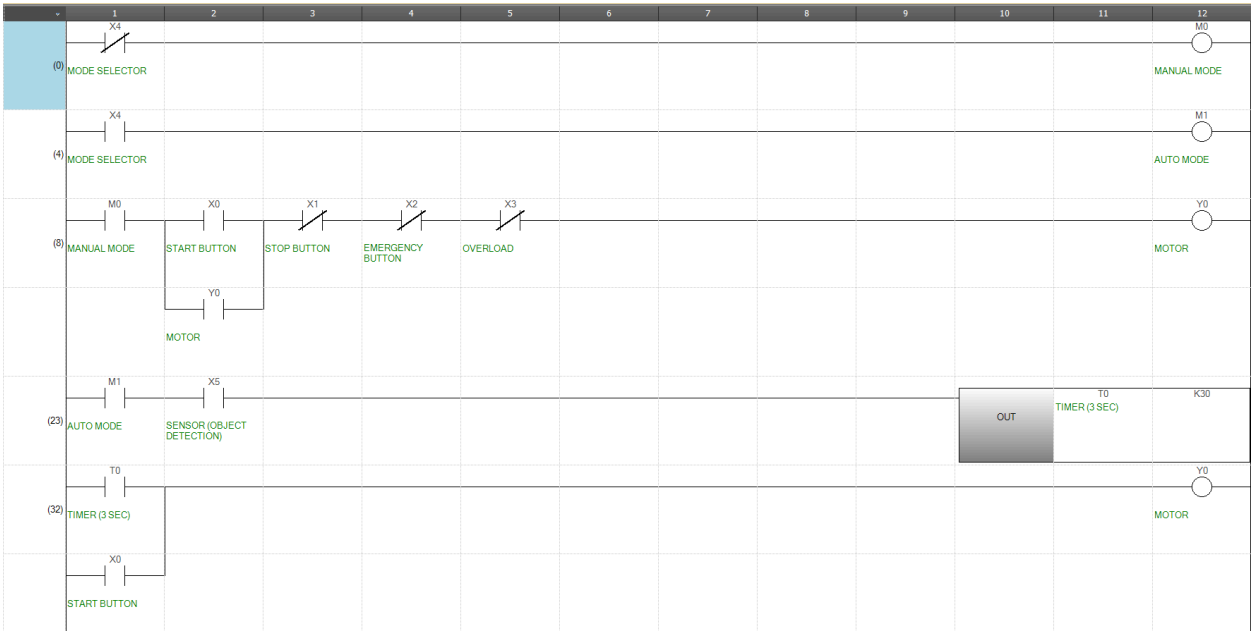
The system performed reliably under all test conditions.

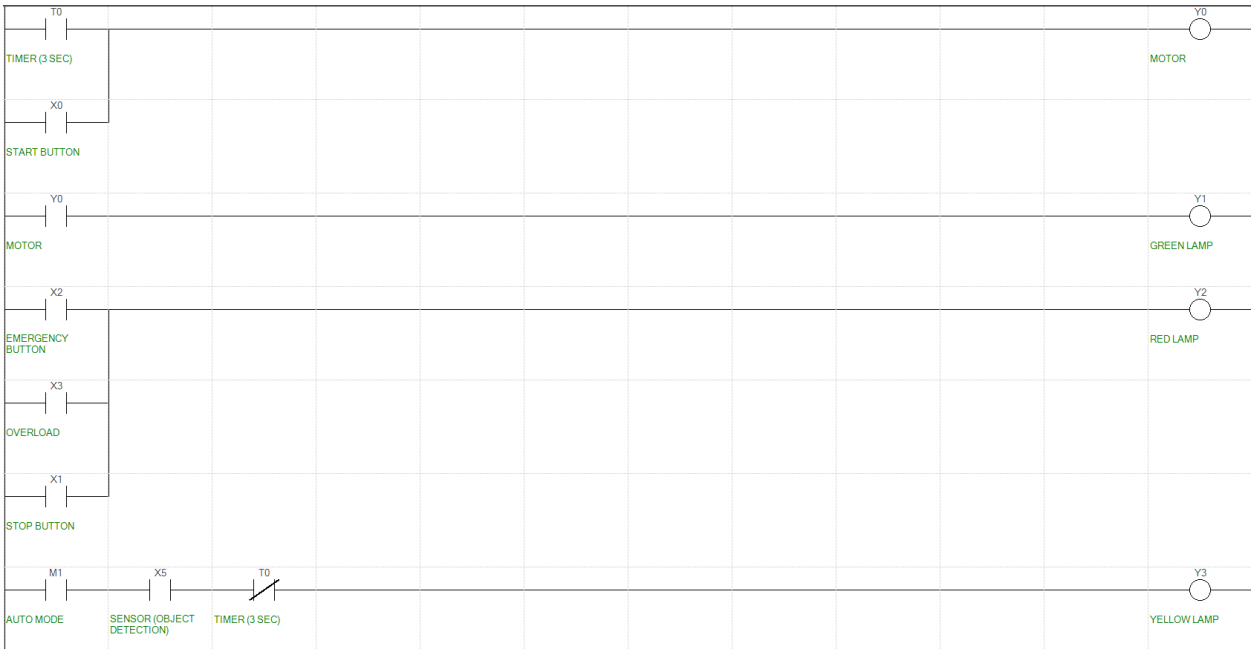
🏁 10. Conclusion

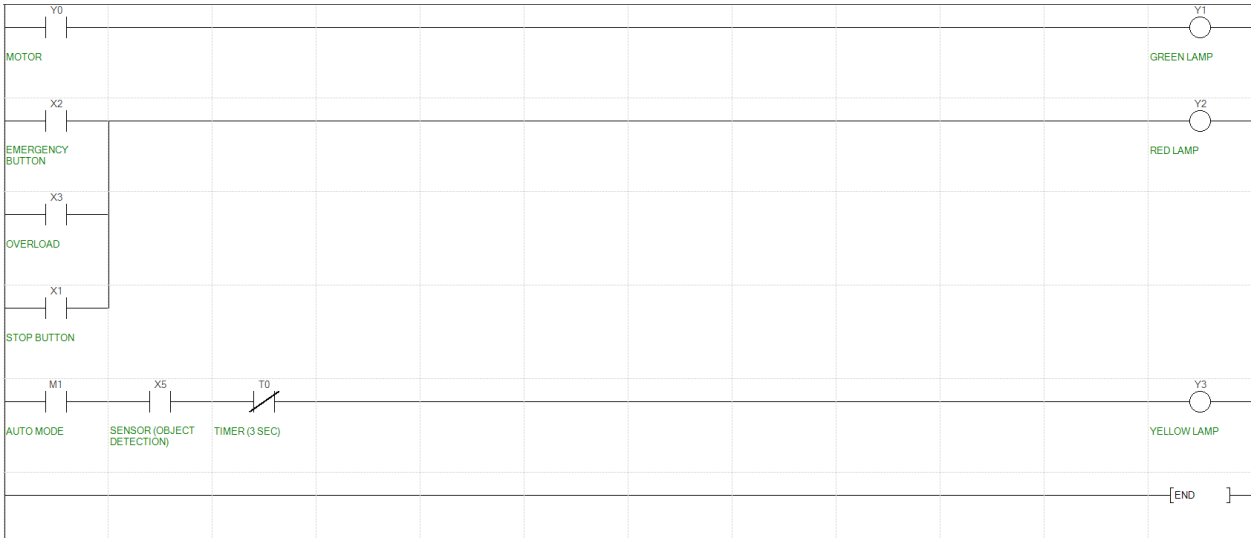
This project successfully demonstrates the design and implementation of a PLC-based conveyor motor control system. The integration of manual and automatic modes, timer-based control, and safety interlocks ensures efficient and safe operation.

The system reflects real industrial automation practices and provides a strong foundation for further development in PLC and automation systems.

Ladder Logic Images




T0
TIMER (3 SEC)
X0
START BUTTON
Y0
MOTOR
Y0
MOTOR
Y1
GREEN LAMP
X2
EMERGENCY BUTTON
X3
OVERLOAD
X1
STOP BUTTON
Y2
RED LAMP
M1
AUTO MODE
X5
SENSOR (OBJECT DETECTION)
T0
TIMER (3 SEC)
Y3
YELLOW LAMP


Y0
MOTOR
Y1
GREEN LAMP
X2
EMERGENCY BUTTON
X3
OVERLOAD
X1
STOP BUTTON
Y2
RED LAMP
M1
AUTO MODE
X5
SENSOR (OBJECT DETECTION)
T0
TIMER (3 SEC)
Y3
YELLOW LAMP
END